

Building Our Digital World: Computer Systems and Architecture

COMP1860

School of Computer Science, University of Leeds

Activity Sheet 1.2

This worksheet contains a combination of formative activities and activities that contribute towards your portfolio. Portfolio activities are detailed in the section “Portfolio Task”. Activities marked by (*) are advanced activities as may take some time to complete.

Expectations:

- 1. Timeliness** You should complete all of the activities in the order provided. Activities marked as part of the portfolio should be submitted to Gradescope before the completion date (Friday, 16/10/2026, 14:00).
- 2. Presentation:** You should present all of your work clearly and concisely following any additional guidance provided by the module staff in the module handbook.
- 3. Integrity:** You are responsible that the evidence you submit as part of your portfolio evidence is entirely your own work. You can find out more about Academic integrity on the Skill@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on activities submitted as part of the portfolio will be available on Gradescope.

Support opportunities: Support with the activity sheet is available in the Lab classes and tutorials. Individual support is available via the online booking system.

Expected time for completion: 2-3 hours.

Expected complete date: Friday, 16/10/2026, 14:00

Coursework summary

This worksheet contains exercises on:

- conversion of denary (base 10) numbers to binary and hexadecimal, and vice versa
- binary arithmetic
- representations of integers and floating-point numbers as sequences of bits

These are all topics covered in week 2 of this module. You should refer to the lecture notes and recordings from lectures 1.2.1 and 1.2.2, when attempting these exercises.

Learning outcomes

On completion of this activity sheet you will:

1. Explain the bit-level representations of fixed-length numbers.
2. Translate numerical data from one format to another.

Portfolio Task

The portfolio task is a Gradescope online quiz. You will find a link to it in the COMP1860 Minerva module.

Formative Questions

Binary & Hexadecimal

Conversion

1. Convert the following denary (base 10) numbers to binary (base 2):
 - (a) 3
 - (b) 12
 - (c) 126
 - (d) 128
 - (e) 235
2. Convert the following binary numbers to denary:
 - (a) 10
 - (b) 101
 - (c) 1011
 - (d) 11010
 - (e) 11100101
3. Convert the binary number **10110011** to hexadecimal.
4. Convert the hexadecimal number **A5** to decimal.
5. Convert the decimal number **270** to hexadecimal.

Binary Arithmetic

1. Add the following binary numbers using binary addition:
 - (a) 11 and 10
 - (b) 100 and 110
 - (c) 1101 and 1001
 - (d) 1101 and 10101
 - (e) 10110010 and 10001101
2. Multiply the following binary numbers using binary long multiplication:
 - (a) 11 and 10
 - (b) 111 and 10
 - (c) 1101 and 110

Binary Fractions

1. Convert the following binary fractions to decimal fractions:
 - (a) 1.1
 - (b) 10.01
 - (c) 111.11
 - (d) 101.101
 - (e) 1011.111
2. Convert the following decimal fractions to binary fractions:
 - (a) 1.5
 - (b) 22.25
 - (c) 23.75
 - (d) 25.125
3. (*) Can the decimal floating-point number 3.2 be converted to binary? *Reflection:* If not, how might this be approximated in binary?

Integer & Floating-point Representation

Integer Representation

1. What is the binary representation of **125** in 4-bit binary coded decimal format?
2. What is the binary representation of **125** in 8-bit two's complement integer format?
3. What is the binary representation of **-125** in 8-bit two's complement integer format?
4. (*) For the decimal numbers 72 and -56:
 - (a) Convert each number to 8-bit two's complement integer format.
 - (b) Add the two resulting numbers using binary addition. **Note:** Remember to adjust the result if it is longer than 8 bits.
 - (c) Convert your result back to decimal. Is this correct?

IEEE 754 Floating-point Format

1. (*) Convert the following decimal numbers to single precision 32-bit IEEE 754 floating point format:
 - (a) 235.5
 - (b) 1024.25
 - (c) 12345.375